

Deep Residual Metric Learning for Human Re-identification in Video Surveillance-based Affective Computing

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Abstract

In this study, the recently popular metric learning problem in human re-identification for affective computing is emphasized, and a novel method inspired by the well-known deep residual networks is proposed for the first time. This new method is composed of two parts. The first part is a DRN part with a structure of twin-deep residual networks, each of which has three-branch deep residual network equipped with the conventional deep neural network as its essential non-linear mapping element. The second part is the Mahalanobis metric learning layer. This new method is trained through end-to-end learning, and the Mahalanobis metric as well as latent features from the DRN part could be determined automatically. Experimental analysis based on the CUHK01 database suggests that this new method outperforms several recently proposed deep learning methods and conventional shallow learning methods of metric learning for human re-identification in affective computing.

Index Terms: deep residual network, metric learning, human re-identification, affective computing

1. Introduction

Affective computing is an emerging field in contemporary multimedia studies. When the massive and heterogeneous media metadata are incorporated, affective computing can be associated with multidisciplinary studies, which will in turn be benefited by affective computing. For instance, when the conventional web-based data mining task is tackled together with affective computing, emotional analysis can be fulfilled afterwards to reveal the actual interests and potential behaviors of ending users. When visual metadata (i.e., video clips, images, etc) is focused in video surveillance-based affective computing, it becomes valuable to find out the intuition of pedestrians after they are localized in video sequences. In this study, an interdisciplinary study of affective computing is introduced. The prerequisite step in video surveillance-based affective computing, i.e., the human re-identification problem, is concentrated. Generally speaking, in order to realize affective computing based on visual metadata, the human target needs to be first accurately located within each individual frame of the video sequence captured either through single-camera view or multi-views, before actual emotional analysis can be conducted on the identified human target. Moreover, to fulfill human re-identification in video surveillance-based affective computing, the key metric learning problem is emphasized in this study. Generally speaking, in psychology, there is an important theoretical assumption stating that, the similarity between two percepts (i.e., objects) is

inversely proportional towards the pairwise psychological distance between them. Hence, the more two objects are far away from each other, the less similar they are supposed to be. Therefore, the notion of similarity is closely related to the concept of distance. For metric, it is a special form of distance with particular characteristics. Usually, metric-based similarity or metric itself is often of great importance in machine learning studies. For example, in the popular k -nearest-neighbor (KNN) classifier, the main idea is to classify an object into a group, which is the most common one among groups of its k nearest objects. Metric / metric-based similarity in KNN is often used to indicate those nearest objects, and the performance of KNN is known to be highly influenced by such an incorporated metric / metric-based similarity. In general image retrieval studies, it is also widely accepted that the metric / metric-based similarity plays a crucial part since it helps to determine the degree of relevance of retrieved images to the query.

In this study, a novel method inspired by the recently proposed deep residual networks [1] is proposed for the first time to realize metric learning in human re-identification for video surveillance-based affective computing. This new method is composed of two parts. The first part is with a structure of twin-deep residual networks, each of which has three-branch deep residual network equipped with the conventional deep neural network as its essential non-linear mapping function. The second part is a Mahalanobis metric learning layer. This new method is trained through end-to-end learning, and the Mahalanobis metric as well as latent features from the deep residual networks part could be determined automatically. The structure of this paper is as follows. In Section 2, related studies in deep learning / shallow learning-based metric learning are introduced with representative studies highlighted. In Section 3, technical details of the new method are elaborated. The structure and the training strategy of the new method are introduced in detail. In Section 4, experimental analysis based on the well-known CUHK01 database is conducted. This new method is also compared with several recently proposed deep learning methods as well as conventional shallow learning methods. Statistical analysis is also conducted to reveal the superiority of the new method. In Section 5, the conclusion of this study is drawn and potential future directions are suggested therein.

2. Related Work

Metric / metric-based similarity as well as their representative learning studies are reviewed in this section. In subsection 2.1, the concept of metric / metric-based similarity is the focus.

Characteristics of metrics are presented with several popular metrics introduced. In subsection 2.2, popular metric / metric-based similarity learning methods proposed from both the conventional shallow learning perspective and the currently popular deep learning point of view are introduced and discussed.

2.1. Metric and Metric-based Similarity

Metric is a special type of distance with several unique characteristics satisfied. Given three objects X , Y , and Z ; D denotes the distance between any two pairwise objects. In order to make D a metric, four axioms must be satisfied simultaneously, and they are non-negativity (i.e., $D(X, Y) \geq 0$), identity of indiscernibles (i.e., $D(X, Y) = 0$, iff $X = Y$), symmetry (i.e., $D(X, Y) = D(Y, X)$), and triangle inequality (i.e., $D(X, Y) \leq D(X, Z) + D(Y, Z)$). Violating one or more axioms can result in the named generalized metric, which in fact does not strictly follow all four requirements of becoming a metric, but can be considered as a special form of metric. For example, if the axiom of “identity of indiscernibles” cannot be fully met (i.e. the condition “iff”, which means if and only if, is replaced as “if”), the resulted generalized metric is a pseudo-metric. If the axiom of “symmetry” is discarded, the resulted generalized metric is a quasi-metric (i.e., one thing to take note here is that, a quasi-metric can become a real metric by simply combining two related quasi-metrics together: $\hat{D}(X, Y) = \frac{1}{2}(D(X, Y) + D(Y, X))$). If the axiom of “triangle inequality” cannot be fulfilled, the resulted generalized metric is a semi-metric. To sum up, distance does not need to follow all four axioms of being a metric. Thus, metric becomes a special form of distance.

There are a number of popular metrics proposed in literature. Given x and y are d -dimensional vectors representing objects X and Y respectively, one of the most influential metrics in machine learning and pattern recognition is the Euclidean metric (a.k.a. L_2 metric), which can be represented as: $D(X, Y) = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$. Euclidean metric belongs to the big family of Minkowski metric (a.k.a. L_p metric), which is often written via a generalized form $L_p(X, Y) = (\sum_{i=1}^d |x_i - y_i|^p)^{\frac{1}{p}}$ (when $p = 2$, $L_p(X, Y)$ becomes $L_2(X, Y)$, which is the Euclidean metric). Furthermore, Minkowski metric can be used to generate two other well-known types of metrics, including the Manhattan metric (i.e. $p = 1$, $L_1(X, Y) = \sum_{i=1}^d |x_i - y_i|$) and the Chebyshev metric (i.e. $p = \infty$, $L_\infty(X, Y) = \max |x_i - y_i|, i = 1, \dots, d$). Another widely used metric is the Mahalanobis metric, which can be represented as $D(X, Y) = \sqrt{(x - y)A(x - y)^T}$, where A is related to the co-variance matrix of data. For Mahalanobis metric, in order to fulfill the four axioms of a metric, the matrix A needs to be at least positive semi-definite (PSD), viz. $A \succeq 0$. The explanation is straightforward. If A is not PSD, suppose $v = x - y$ is a normalized eigenvector corresponding to a negative eigenvalue λ of A , the corresponding Mahalanobis metric could be negative: $(x - y)A(x - y)^T = vAv^T = \lambda vv^T = \lambda < 0$, which obviously violate the “non-negativity” axiom. However, in order to make it a strick metric, A needs to be positive definite, viz. $A \succ 0$. The reason here is because, the sufficient condition of $X = Y$ in the axiom of “identity of indiscernibles” for this case could be violated when $A = 0$, and the metric is a pseudo-metric in this case. However, in most existing metric learning studies, a pseudo-metric is also regarded as a metric with loose definitions. Hence, for Mahalanobis metric, the matrix A can be considered as PSD for most

cases.

There are also other types of metrics proposed from diverse points of view. Arising from the probability and information theory, Kullback-Leibler (KL) divergence, which is a quasi-metric, can be utilized to reveal the difference between two probability distributions P and Q with a definition: $KL(P, Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$. In order to make it a real metric, Jensen-Shannon divergence is proposed to overcome the asymmetric problem of KL divergence by combining two KL divergences together: $JS(P, Q) = \sqrt{\frac{1}{2}KL(P, M) + \frac{1}{2}KL(Q, M)}$, in which $M = \frac{1}{2}(P + Q)$. Another popular metric arising from the probability theory is Earth Mover’s Distance (i.e., EMD, also known as one type of the Wasserstein metric), which also measures the distance between two probability distributions over a specific region. The main idea of EMD is to calculate the minimal cost of transforming one distribution to another with the help of a defined “signature” (i.e. a descriptor of extracted clusters from original distributions together with weights indicating the size of each cluster) and solved as a typical linear programming problem.

Having introduced the above popular metrics, metric-based similarity can be generated from them through various manners. Generally speaking, two requirements should to be fulfilled in the process, including 1) metric-based similarity should be inverse proportional towards its corresponding metric, and 2) values of the generated similarity between two objects should be non-negative in the common sense. In this study, the utilized metric-based similarities are constructed via a simple but effective way based on an exponential form of the inverse metric (i.e. $\exp(-D(X, Y))$), so that the inverse relationship between metric and similarity as well as the non-negative requirement of similarity can be fulfilled simultaneously.

2.2. Metric / Metric-based Similarity Learning

Based on the above description, it is also clear that metric-based similarity learning is equivalent towards explicitly learning a metric. Based on the availability of supervisory information within the training data, metric learning methods can be mainly categorized into three kinds. One is supervised learning, which is mainly utilized in classification problems. Supervised similarity learning methods are usually optimization-based. In practice, a parametric form of the utilized metric is often explicitly defined, and its optimal parameters are learned by optimizing a criterion function [2]. Another kind of learning method is unsupervised learning, which can be used for clustering problems [3]. Also, an alternative learning trend known as semi-supervised learning, which is in-between the supervised and unsupervised learning paradigms with the help of some additional supervisory information, receives much research attention as well [4]. The additional supervisory information used in semi-supervised learning methods can be further divided into two types. The first one is composed of limited labeled data. The number of these labeled examples is quite small compared with that of unlabeled examples. The second type of information is in the form of additional constraints, in which, side-information represented by two pairwise constraints is one of the most popular [5]. It is also necessary to mention that, side-information is often weaker than limited labeled data, as pairwise constraints can be derived from labeled data but not vice versa.

Accompanying with the recently popular deep learning techniques, more and more metric learning studies are proposed from the perspective of deep learning nowadays. [6]–[20] sum-

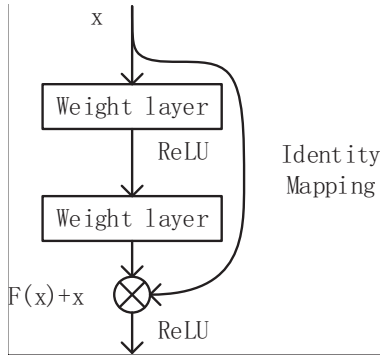


Figure 1: An illustration of making stacked layers fit a residual mapping in the original deep residual network model (originally illustrated in [1], small revisions are applied here)

marize representative papers searched from the web-of-science database using the keyword “deep metric learning”. It can be concluded from these related studies that, the number of deep metric learning studies in recent two years [8]–[20] demonstrates a significant increase compared with that before [6]–[7], which suggests that deep metric learning is one of the currently popular trends in metric learning and deep learning. Also, the recently popular deep learning techniques, including convolutional neural network [6]–[7], deep neural network [9], deep belief network [11]–[15], auto-encoder [18]–[20], etc, have been employed to bring about diverse deep metric learning models, respectively. When considering the application domain, it has been expanding from the conventional tasks of person identification [6]–[11] to others including pedestrian tracking [14], image retrieval [16], signature verification [17], 3D modeling [18], behavior recognition [20], etc, in the last two years. Also, most of these deep metric learning studies still heavily rely on conventional techniques commonly utilized in machine learning, including regularization, gradient descent and its variations, etc, to fulfill automatic tuning of unknown parameters in their metrics [6]–[18], [20]. Therefore, the above recent progress in deep metric learning inspires us to introduce our new method based on one of the state-of-the-art deep learning techniques, i.e., deep residual networks, to realize the metric learning task. Details of the new method are elaborated in Section 3.

3. Methodology

3.1. Deep Residual Network

The original single-channel deep residual network (ResNet) was initially introduced in the year 2015 [1], and becomes widely acknowledged as one of the state-of-the-art deep learning models for the time being, provided the fact that this model dominates all five major tracks of the noticeable ImageNet and COCO competitions when it was proposed [1]. Its successor also won the first place of the detection track in the ImageNet competition in the year 2016. The motivation of ResNet is to tackle the notorious degradation problem, which implies that the accuracy of a deep network will get saturated and then degrades rapidly with the increase of the network depth. A simple solution to solve the degeneration problem is to design a shallower architecture of the deep network. However, the generalization capability of the shallower network will certainly be deteriorated, and a deeper structure of the network is always preferred for its attractive potentiality in representing more sophis-

ticated non-linear relationship and guaranteeing more promising generalization capability. ResNet provides an ideal solution to overcome the above dilemma by allowing a tremendously deep structure (i.e., as deep as 152 layers described in [1]) while achieving accuracy gain from considerably increased network depth. The key point of ResNet is to make stacked layers fit a residual mapping, instead of a desired underlying mapping in other deep learning studies. An illustration of the residual mapping is displayed in Figure 1, where X represents the input of the stacked layers (i.e., two weighted layers for an illustration purpose in Figure 1), and the output is the sum of the weighted mapping $F(X)$ and an identity mapping X . The reason to conduct the residual mapping instead of the desired mapping is because it is believed to be more convenient in optimizations [1]. It is also necessary to mention that, the stacked weighted mapping $F(X)$ can be explicitly defined in many ways, and it is in the form of the conventional deep convolutional neural network (CNN) in the original single-channel ResNet [1].

3.2. Deep Residual Metric Learning

In this study, a novel deep residual metric learning (DRML) method inspired by the original ResNet is introduced for human re-identification in video surveillance-based affective computing, for the first time. Generally speaking, the main task of human re-identification is to discern the identity of one unknown human, and a suitable metric is necessary here to allocate the lowest measurement between the testing image of this unknown human and her/his another image available in a database. Meanwhile, images between different human should be revealed via large measurements computed from this metric. The structure of the new DRML method is described in Figure 2. There are two main components within this structure, including a deep residual network (DRN) part and a following Mahalanobis metric layer part. The first DRN part contains twin-deep residual networks whose weights are shared. After latent features are determined from the DRN part, the following Mahalanobis metric layer is employed for learning a suitable Mahalanobis afterwards. The training of the whole structure can be back-propagated via the gradient method, in order to fulfill an end-to-end learning of this deep learning method. It is necessary to point out that, the DRN part of Figure 2, in which a novel three-branch structure resides, is firstly proposed and reveals the technical novelty of this study.

3.2.1. Training of the DRN Part

The training strategy of the DRN part in DRML is described in this subsection. For each individual one of the twin-deep residual networks, there are actually three branches as illustrated in Figure 3. First, the spacial resolution of an input color image is pre-defined as 128×64 , and a combination of 3 overlapping image patches whose spacial resolution is set as 64×64 is obtained from this original input image. Each individual patch is then fed into its corresponding branch of the three-branch model. Moreover, each branch is composed of a typical ResNet, with which CNNs are mainly equipped. One thing to take note here is that, the weights of CNNs among three branches cannot be shared, as they have to be determined via different patches fed into each individual branch. Also, in each branch, CNNs actually act as the weighted mapping $F(X)$ in Figure 1 and a convolution kernel of the size 1×1 is employed to realize the shortcut as an identity mapping (illustrated in Figure 4). In this way, the basic idea of ResNet in Figure 1 can be fulfilled in both each branch (Figure 3) and then the whole DRML (Figure 2). After that, the

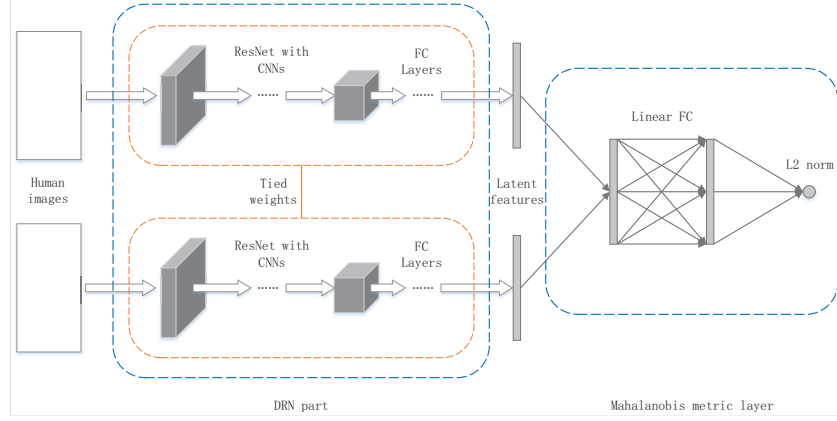


Figure 2: The structure of the new deep residual metric learning (DRML) method for human identification in video surveillance-based affective computing

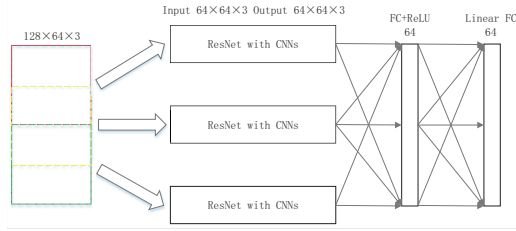


Figure 3: Details of the three-branch deep residual network in DRN part)

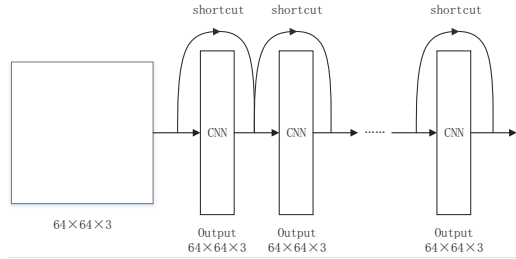


Figure 4: ResNet with CNNs in each branch of the three-branch model)

output of the three-branch model is directly connected towards a full connection (FC) layer with rectified linear units (ReLU) as activation functions. Then, a linear FC layer is incorporated, the trained latent feature can be extracted afterwards and utilized as the input of the following Mahalanobis metric layer.

3.2.2. Mahalanobis Metric Learning

The Mahalanobis metric is to be learned via DRML in this study. Provided τ_1 and τ_2 as images sequences obtained from two views in human re-identification, X_1 and X_2 are latent features obtained from the DRN part of the two images sequences, respectively. Given $x_1 \in X_1$ and $x_2^p \in X_2$ as positive pairs (i.e., meaning they actually represent the same human); $x_1 \in X_1$ and $x_2^n \in X_2$ as negative pairs (i.e., meaning they indicate different human), the purpose of the Mahalanobis metric is to allocate small metric values towards positive pairs while identifying negative pairs through large metric values. The ex-

plicit form of the Mahalanobis metric is as Equation 1.

$$d(x_1, x_2) = \sqrt{(x_1 - x_2)^T M (x_1 - x_2)} \quad (1)$$

where, $x_2 \in \{x_2^p, x_2^n\}$ and M is the co-variance matrix to be learned. In DRML, x_1 and x_2 are latent features obtained from the DRN part, while the weights of DRML as well as the co-variance matrix M are to be determined via back-propagation. Normally, the co-variance matrix M needs to be symmetric and PSD, to ensure requirements of a metric to be met. Thus, M can be further decomposed as $M = WW^T$ via Cholesky factorization. Hence, Equation 1 can be further derived as Equation 2.

$$\begin{aligned} d(x_1, x_2) &= \sqrt{(x_1 - x_2)^T WW^T (x_1 - x_2)} \\ &= \sqrt{(W^T (x_1 - x_2))^T (W^T (x_1 - x_2))} \\ &= \|W^T (x_1 - x_2)\|_2 \end{aligned} \quad (2)$$

in which, $W^T (x_1 - x_2)$ can be realized via a linear FC illustrated in the Mahalanobis metric layer in Figure 2. For its implementation, the above metric is fed into an energy function $L = \|d(x_1, x_2^p) - d(x_1, x_2^n)\|_2$ and a gradient method is applied to back-propagate for realizing metric learning.

4. Experiments

4.1. Database and Experimental Settings

In order to evaluate the performance of the newly proposed DRML method for human re-identification in video surveillance-based affective computing, the well-known CUHK01 database [21] is incorporated for experimental analysis. For the database, it contains 971 persons, each of whom has 4 images. Therefore, totally 3884 images are included in this database. Besides the new DRML method, there are 9 deep learning-based / shallow learning-based metric learning methods for comparisons. The 3 Deep learning-based methods include DML (i.e., the same steps as DRML but without employing ResNet), Norm+X-Corr [22], and Fused [22], while the rest 6 shallow learning-based methods incorporate gray-level co-occurrence matrix (GLCM) as the low-level hand-crafted features and they are GLCM+SVM, BPNN, GLCM+DCA [23], GLCM+RCA [24], GLCM+NCA [25], and GLCM+LMNN [26].

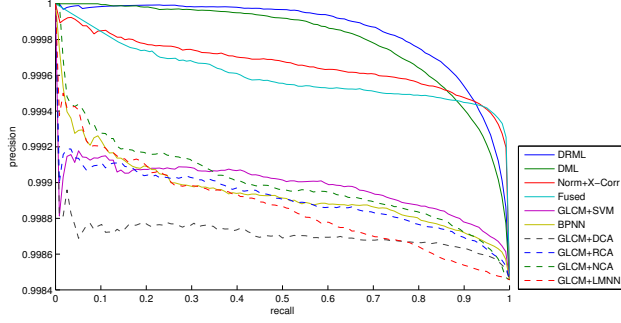


Figure 5: Precision-recall curves of all compared methods in the CUHK01 database

Table 1: Multiple comparison test results of compared deep learning-based methods based on *F*-measure.

Method I	Method II	Single-value Estimation (I-II)	95% Confidence Interval
DRML	DML	0.0839	[-0.0381, 0.2058]
DRML	Norm+X-Corr	0.0839	[-0.0380, 0.2059]
DRML	Fused	0.0839	[-0.0381, 0.2059]

4.2. Experimental Results and Statistical Analysis

In experiments, images from 485 persons of the CUHK01 database are randomly chosen to construct the training dataset, while images from the rest 486 persons are utilized as the testing data. The above process is repeated to alleviate the uncertainty of randomness, and experimental outcomes are obtained based on the average estimation of all experimental outcomes. Precision-recall curves of all compared methods are illustrated in Figure 5. Since the area below each individual precision-recall curve is the mean precision of the corresponding method, it can be easily observed from Figure 5 that, deep learning-based methods demonstrate better performance than shallow learning-based methods for human re-identification in this study. In order to quantitatively evaluate the performance of DRML compared with other deep learning-based methods, *F*-measure is calculated based on obtained precision and recall outcomes (i.e., $F - measure = \frac{2 \times precision \times recall}{precision + recall}$). Two ways of statistical estimation, including single-value estimation and 95% confidence interval estimation, are conducted based on all calculated *F*-measure outcomes. Their details are listed in Table 1. It can be noticed that, the single-value estimation results are all positive. Since this estimation is calculated as the direct mean difference between Method I (i.e., DRML) and Method II (i.e., other compared deep learning-based methods), it is a clear sign that DRML outperforms others regarding the single-value estimation on *F*-measure. Also, the upper bound of all 95% confidence intervals in Table 1 are larger than the magnitude of their corresponding low bounds, which also indicates that although compared deep learning-based methods can perform well in some cases, the majority of cases in the CUHK01 database are still dominated by DRML. Hence, it can be concluded from Figure 5 and Table 1 that, deep learning-based methods can provide better performance than compared shallow-learning methods in this metric learning study for video surveillance-based human re-identification. Among all compared deep learning-based methods, DRML can

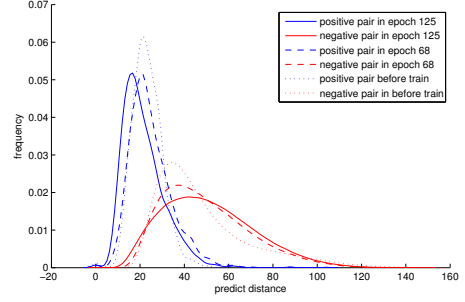


Figure 6: Distributions of positive pairs and negative pairs of testing data with different settings of training epochs in DRML

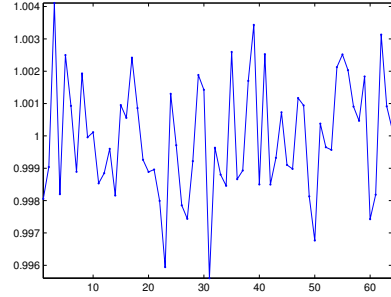


Figure 7: Learned diagonal elements of the co-variance matrix W of the Mahalanobis metric obtained by DRML in this study

outperform compared recently proposed ones (i.e., Norm+X-Corr and Fused), while the superiority of adopting ResNet in DRML is also demonstrated by the comparison between DRML and DML in Table 1.

4.3. Discussions

In this subsection, more technical details about DRML are discussed. First, Figure 6 displays distributions of positive pairs and negative pairs of testing data with different settings of training epochs in DRML. It can be noticed that, with the increase of training epochs, the difference between positive pairs and negative pairs of testing data becomes larger. It suggests that the generalization capability of DRML could be guaranteed via comprehensive training of the deep learning model, although the computational time could be increased by leaps and bounds with the increase of training epochs. Therefore, a good balance between computational efficiency and experimental effectiveness needs to be properly determined. Second, Figure 7 demonstrates learned diagonal elements of the co-variance matrix W of the Mahalanobis metric obtained by DRML in this study. For the rest non-diagonal elements of W , most of them are nearly zero. Thus, the correlation between different dimensions of latent features obtained from the DRN part of DRML could be almost neglected, which is a good sign of the end-to-end learning-based feature selection of DRML. In Figure 7, the 3rd dimension results in the largest learned weight (i.e., 1.0040), while the 31st dimension provides the lowest weight (i.e., 0.9956). Third, Figure 8 demonstrates error examples of DRML in this study. To be specific, the upper row of Figure 8 displays the false positive while its lower row demonstrates the false negative. It can



Figure 8: Examples of false positive (upper row) and false negative (lower row) of DRML in this study

be noticed from the false positive that, when the background and the brightness vary, errors are prone to be generated even the same human is spotted. Also, when the gesture of different human is similar, false negative is likely to occur. The above shortcomings of DRML may be alleviated by incorporating specific color space / gesture-based hand-crafted features in the deep learning-based framework through a fusion of shallow learning-based features and deep learning-based features.

5. Conclusion

In this study, a novel method of metric learning inspired by the recently proposed deep residual network is introduced for solving human re-identification in video surveillance-based affective computing, for the first time. Two parts are made up of this new method. The first part is the DRN part, which has a structure of twin-deep residual networks, each of which has three-branch deep residual network with the well-known deep convolution neural network as its essential non-linear mapping element. The second part is the Mahalanobis metric learning layer, whose main task is to determine a proper Mahalanobis metric through end-to-end learnings based on trainable latent features. Experimental analysis based on the well-known CUHK01 database suggests that this new method outperforms several recently proposed deep learning methods and conventional shallow learning methods of metric learning for human re-identification in video surveillance-based affective computing. Future studies will continue with more sophisticated deep learning models equipped with more elegant metric learning algorithms.

6. Acknowledgements

The authors would like to acknowledge grants 61403182, 61363046, 61463032, 61562060, 61301194, and 61231016 approved by National Natural Science Foundation of China, the key youth grant 20171ACB21017 approved by Natural Science Foundation of Jiangxi Province, the provincial grant JXJG-15-1-26 approved by the Jiangxi Province Office of Education, the Natural Science Foundation Grant 827/000088 of SZU, the Research Fund for the Doctoral Program of Higher Education of China 20126102120055, as well as the foundation grant from NWPU 3102014JSJ0014 for supporting this study.

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